Lab 3: Sensors and Reliable Measurement

Photoresistor Calibration and Ultrasonic Distance

ENGR 1105 | Fall 2026

Name	**Partner**
Date	**Section**

Learning Objectives

By the end of this lab, you should be able to:

1. build and test a photoresistor voltage-divider circuit;
2. collect light-sensor readings under known reference conditions;
3. choose a threshold from measured data and explain hysteresis;
4. wire an HC-SR04 ultrasonic distance sensor;
5. calculate distance from echo travel time;
6. identify and reject invalid distance readings;
7. average repeated measurements; and
8. create a multi-zone distance warning system.

1. Lab Organization

Lab 3 is completed in two class meetings:

Part	Main work
Lab 3.1	Photoresistor circuit, light measurements, calibration, threshold control, and hysteresis
Lab 3.2	Ultrasonic wiring, distance calculation, validation, averaging, warning zones, and a design challenge

2. Materials

Item	Purpose
Arduino Uno and USB cable	Controller, power, and program upload
Solderless breadboard and jumper wires	Temporary circuit construction
Photoresistor and $10\,\text{k}\Omega$ **resistor**	Light-sensing voltage divider
HC-SR04 ultrasonic sensor	Noncontact distance measurement
LED and $330\,\Omega$ **resistor**	Visual output
Computer with Arduino IDE	Code editing, upload, and Serial Monitor
Ruler or tape measure	Distance reference
Flashlight or phone light	Light-level reference

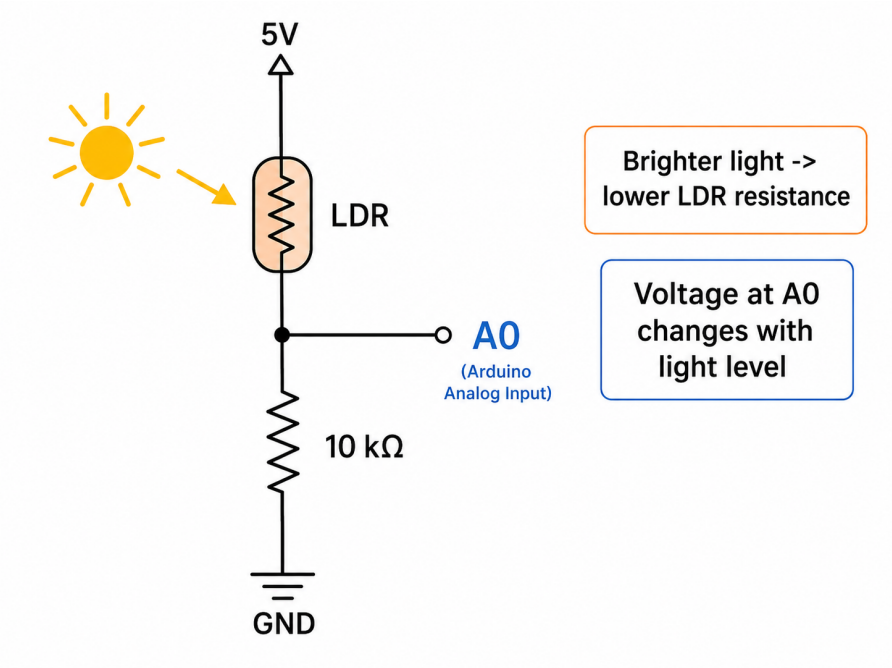
Before Applying Power

1. Disconnect USB power before making major wiring changes.
2. Never connect `5V` directly to `GND`.
3. Use the $10\,\text{k}\Omega$ resistor in the photoresistor divider.
4. Use the $330\,\Omega$ resistor in series with the LED.
5. Check the HC-SR04 pin labels before connecting power.
6. Remove power immediately if the Arduino or a component becomes hot.

Lab 3.1
Photoresistor Calibration and Thresholds
Measuring Light and Creating Stable Decisions

3. Part A: Build the Photoresistor Circuit

A photoresistor, also called an LDR, changes resistance with light. In the circuit below, brighter light normally lowers the LDR resistance and raises the voltage measured at A0.



Wiring Steps

1. Disconnect the Arduino from USB power.
2. Connect one photoresistor lead to Arduino `5V`.
3. Connect the other photoresistor lead to a shared junction row on the breadboard.
4. Connect the 10 kΩ resistor from the junction row to `GND`.
5. Connect the junction row to analog pin A0.
6. Inspect the wiring before reconnecting USB power.

Component Orientation

A photoresistor is not polarized, so either lead may face `5V`. The circuit direction depends on whether the LDR is above or below the fixed resistor. Use the arrangement shown in this handout.

4. Part B: Read the Light Sensor

```
const int lightPin = A0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int lightValue = analogRead(lightPin);
  Serial.println(lightValue);
  delay(100);
}
```

Instructions

1. Upload the program.
2. Open the Serial Monitor and set the baud rate to `9600`.
3. Cover the sensor completely and wait for the reading to stabilize.
4. Record five covered readings.
5. Expose the sensor to normal room light and record five readings.
6. Illuminate the sensor with a flashlight and record five readings.

Light Measurements

Condition	1	2	3	4	5	Average
Covered						
Room light						
Flashlight						

Does brighter light produce a larger or smaller Arduino reading?

What is the approximate useful range between your covered and flashlight averages?

Checkpoint 1: Light Sensor

Demonstrate that the reading changes clearly among covered, room-light, and flashlight conditions. Explain which circuit node is connected to A0.

5. Part C: Choose a Threshold

A threshold separates two operating conditions. Choose a threshold approximately halfway between the covered and room-light averages:

$$\text{threshold} = \frac{\text{covered average} + \text{room-light average}}{2}.$$

Threshold Selection

Covered average: _____ Room-light average: _____

selected threshold = _____

Show the calculation:

Use your measured threshold in the program:

```
const int lightPin = A0;
const int ledPin = 9;
const int lightThreshold = 350;  // replace

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int lightValue = analogRead(lightPin);
  Serial.println(lightValue);

  if (lightValue < lightThreshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
```

```
  }
}
```

Test the Decision

1. Replace 350 with your measured threshold.
2. Upload the program.
3. Cover and uncover the photoresistor.
4. Confirm that the LED changes state.
5. If the behavior is reversed, reverse the comparison sign.

Threshold Results

Condition	Sensor value	LED state
Covered		
Near the threshold		
Bright light		

Why might the LED flicker when the reading is close to the threshold?

Checkpoint 2: Calibrated Threshold

Demonstrate light-controlled LED behavior. Explain how your measured data was used to select the threshold.

6. Part D: Add Hysteresis

Use two thresholds so the LED does not repeatedly switch near one boundary.

```
const int lightPin = A0;
const int ledPin = 9;

const int turnOnThreshold = 330;   // replace
const int turnOffThreshold = 390;  // replace
bool ledOn = false;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  int lightValue = analogRead(lightPin);

  if (!ledOn && lightValue < turnOnThreshold) {
    ledOn = true;
  }

  if (ledOn && lightValue > turnOffThreshold) {
    ledOn = false;
  }

  digitalWrite(ledPin, ledOn ? HIGH : LOW);
}
```

Choose two thresholds separated by approximately 30–60 counts. Adjust the comparison directions if your circuit responds in the opposite direction.

Hysteresis Test

Turn-on threshold: _____ Turn-off threshold: _____

How does the two-threshold behavior differ from the single-threshold behavior?

_____

_____

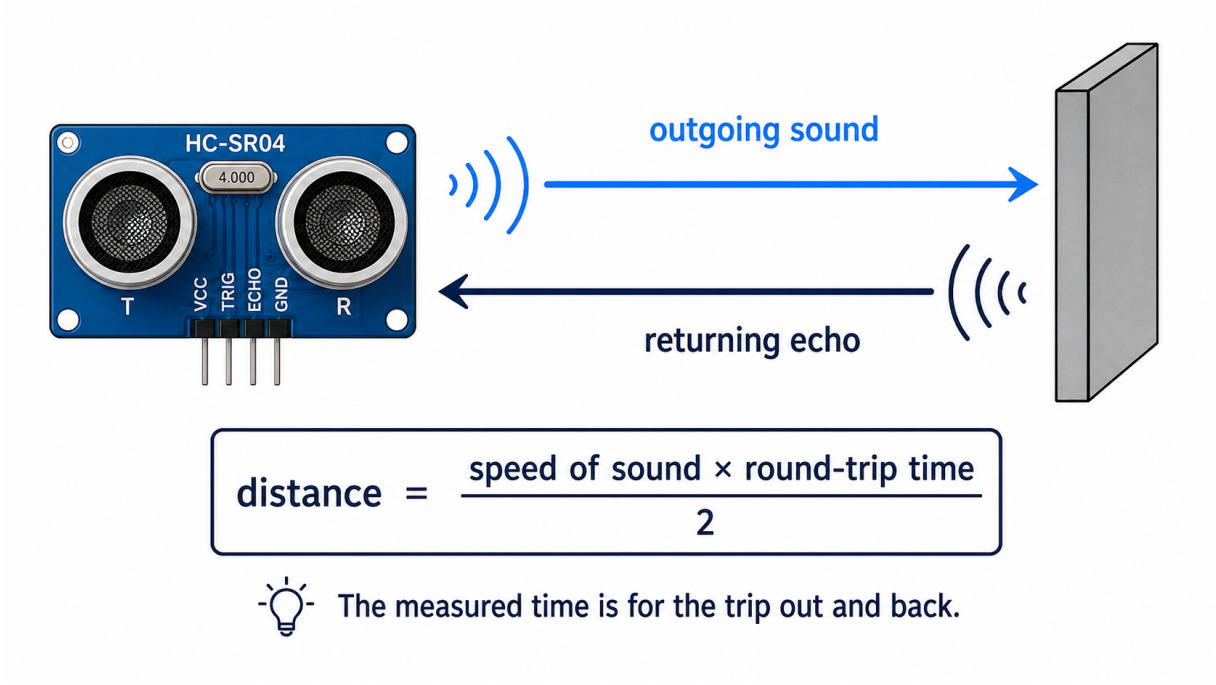
_____

Lab 3.2

Ultrasonic Distance and Reliable Decisions

Time of Flight, Validation, Averaging, and Warning Zones

7. Part E: Wire the HC-SR04 Sensor



HC-SR04 pin	Purpose	Arduino connection
VCC	Sensor power	`5V`
GND	Electrical reference	`GND`
TRIG	Trigger pulse from Arduino	digital pin 7
ECHO	Return pulse to Arduino	digital pin 6

Wiring Steps

1. Disconnect USB power.
2. Identify the pin labels printed on the sensor module.
3. Connect VCC to `5V` and GND to `GND`.
4. Connect TRIG to digital pin 7.

5. Connect ECHO to digital pin 6.
6. Place a large flat object approximately 30 cm in front of the sensor.
7. Inspect the wiring before reconnecting power.

8. Part F: Measure Distance

```
const int trigPin = 7;
const int echoPin = 6;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  unsigned long duration =
      pulseIn(echoPin, HIGH, 30000);

  if (duration == 0) {
    Serial.println("No valid echo");
  } else {
    float distanceCm =
        duration * 0.0343 / 2.0;
    Serial.println(distanceCm);
  }

  delay(100);
}
```

Instructions

1. Upload the program and open the Serial Monitor.
2. Place a large flat target at approximately 10 cm, 25 cm, and 50 cm.
3. Use a ruler to measure the actual distance from the sensor face to the target.
4. Record five Arduino measurements at each reference distance.

Distance Measurements

Reference	1	2	3	4	5	Average
10 cm						
25 cm						
50 cm						

For one test distance, calculate the error:

$$\text{error} = \text{measured average} - \text{reference distance}.$$

Reference distance: _____ Measured average: _____

error = _____

Why is the echo travel distance divided by two?

_____

_____

Checkpoint 3: Distance Measurement

Demonstrate that the reported distance changes as the target moves. Explain the trigger pulse, echo pulse, and divide-by-two calculation.

9. Part G: Test Unreliable Targets

Compare the following targets at approximately the same distance:

Target	Expected measurement behavior
Large flat book facing the sensor	usually stable and reliable
Narrow object or pencil	may be missed or inconsistent
Soft towel or fabric	may absorb or scatter sound
Book held at an angle	may reflect sound away from the receiver

Target Comparison

Target	**Typical reading**	**Stable or unstable?**
Flat book		
Narrow object		
Soft object		
Angled book		

Which target produced the most reliable reading, and why?

10. Part H: Average Valid Readings

Use a function that collects five valid measurements and returns their average.

```
float averageDistance(int samples) {
  float sum = 0.0;
  int validCount = 0;

  for (int i = 0; i < samples; i++) {
    float d = readDistanceCm();
    if (d >= 2.0 && d <= 400.0) {
      sum += d;
      validCount++;
    }
    delay(30);
  }

  if (validCount == 0) return -1.0;
  return sum / validCount;
}
```

Averaging Results

Hold a flat target at approximately 30 cm.

Method	Smallest reading	Largest reading
Single measurements		
Five-sample average		

Which method varied less?

What is one disadvantage of averaging more samples?

11. Part I: Build a Distance Warning System

Keep the photoresistor circuit disconnected. Add an LED to PWM pin 9 through a 330 Ω resistor.

```
const int ledPin = 9;

void loop() {
  float distanceCm = averageDistance(5);

  if (distanceCm < 0) {
    analogWrite(ledPin, 0);
  }
  else if (distanceCm < 15) {
    analogWrite(ledPin, 255);
  }
  else if (distanceCm < 35) {
    analogWrite(ledPin, 100);
  }
  else {
    analogWrite(ledPin, 0);
  }
}
```

Warning-Zone Results

Distance region	PWM command	LED appearance
Below 15 cm	255	
15 cm to below 35 cm	100	
35 cm and farther	0	
Invalid reading	0	

Checkpoint 4: Distance Warning System

Demonstrate the danger, warning, and safe regions. Explain why an invalid reading should not be treated as an object at zero distance.

12. Part J: Design Challenge

Design Challenge

Choose one challenge:

1. **Adjustable warning boundary:** Use the photoresistor reading to select between two distance-warning thresholds.
2. **Blink-rate warning:** Make the LED blink faster as an object approaches the ultrasonic sensor.
3. **Stable alarm:** Add hysteresis so the danger alarm turns on below 18 cm and turns off above 22 cm.
4. **Repeated confirmation:** Require three consecutive close readings before activating the danger state.

Design Record

Which challenge did you choose?

Describe the program changes.

Record one test condition and the resulting behavior.

What problem did you encounter, and how did you troubleshoot it?

13. Final Reflection

Explain how Lab 3 demonstrates the sense-decide-act model. Include one example of measurement validation or filtering that made the decision more reliable.

14. Submission Checklist

☐ Completed Lab 3 instruction sheet

☐ Photoresistor calibration measurements and selected threshold

☐ Ultrasonic reference-distance measurements

☐ Averaging and target-comparison results

☐ Clear photo of the final circuit

☐ Final Arduino code

☐ Completed design challenge and reflection